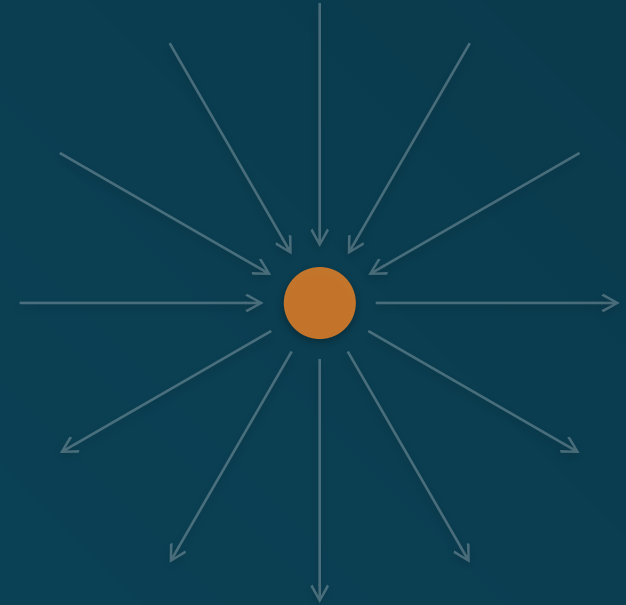


TOPIC 18

Electric Fields

电场



CIE 9702 · A Level Physics A2 · Paper 4

剑桥国际物理 A2 分章节讲义 · 中英双语

Learning Outcomes

学习目标

LEARNING OUTCOMES

学习目标

By the end of this topic, you will
be able to...

18.1 Electric fields & field lines 电场与电场线

- Electric field = force per unit positive charge: $E = F/q$ 场强定义
- Recall and use $F = qE$; represent fields by field lines 电场线表示

18.2 Uniform electric fields 匀强电场

- $E = \Delta V/\Delta d$ between charged parallel plates 平行板场强
- Effect of a uniform field on moving charged particles 带电粒子偏转

18.3–18.4 Point charges 点电荷

- Coulomb's law $F = Q_1Q_2/(4\pi\epsilon_0r^2)$ 库仑定律
- Field of a point charge $E = Q/(4\pi\epsilon_0r^2)$ 点电荷场强

18.5 Electric potential 电势

- Potential $V = Q/(4\pi\epsilon_0r)$; potential energy $E_p = qV$ 电势与电势能

Electric Field & Field Lines

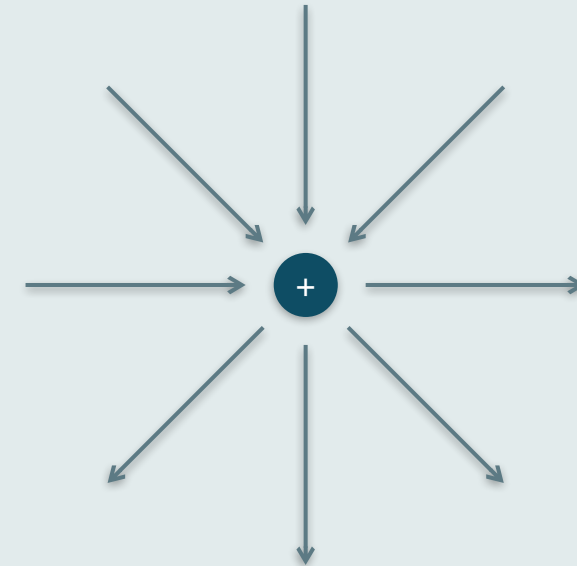
电场与电场线

An **electric field** is a region of space where a **stationary charge** experiences a force. Field strength E is the **force per unit positive charge**.

电场是电荷受力的区域；场强 = 单位正电荷受力。

$$E = \frac{F}{q} \quad F = qE \quad \text{矢量} \cdot \text{单位 } \text{N C}^{-1} (= \text{V m}^{-1})$$

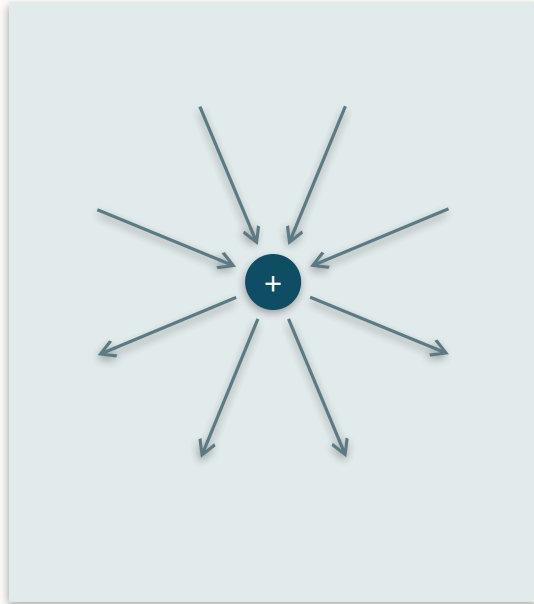
- Field lines point from + to – (正电荷出发，负电荷终止)
- Line density shows field strength 疏密表强弱；lines never cross 不相交
- Direction = force on a positive test charge 方向 = 正电荷受力方向



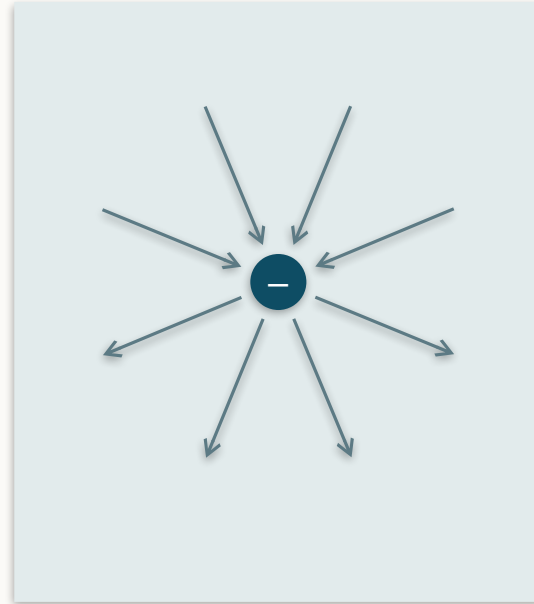
field lines of a point charge 电荷点 charge 电荷的场线

Field Patterns

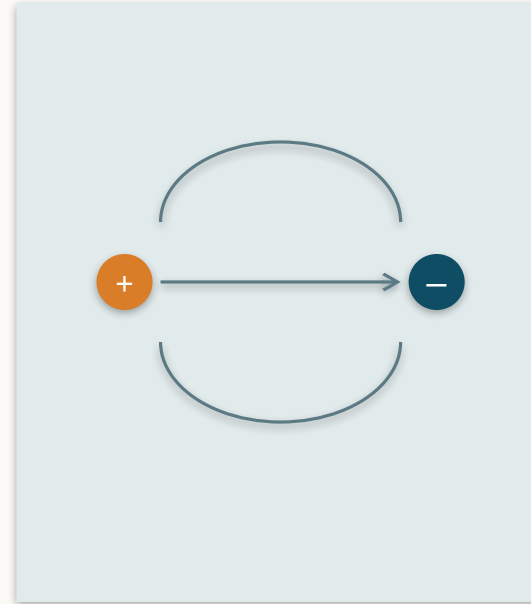
典型电场线分布



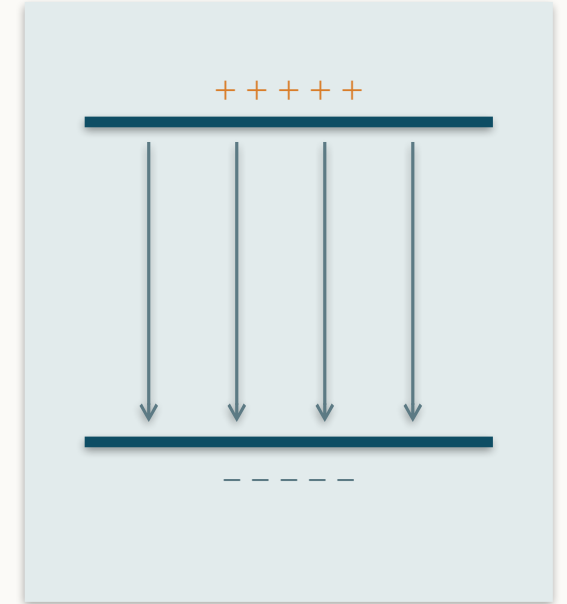
Positive point charge 正点电荷



Negative point charge 负点电荷



Opposite charges 异种电荷



Parallel plates 平行板

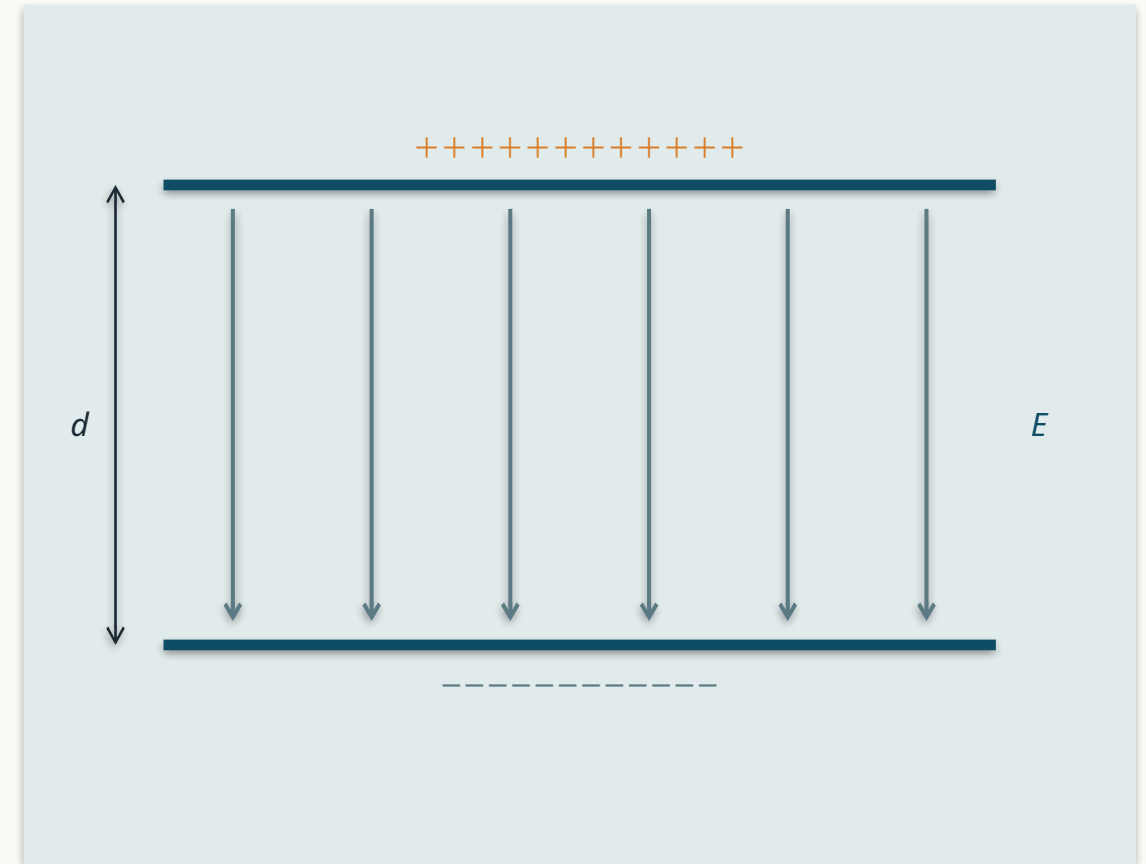
Rules 规则: lines leave + and end on - ; density $\propto$ strength 密 = 强; never cross 不相交; parallel plates give a **uniform** field (edge effects ignored 忽略边缘)。

Uniform Fields

平行板间的匀强电场

$$E = \frac{\Delta V}{\Delta d} \quad \text{匀强电场场强}$$

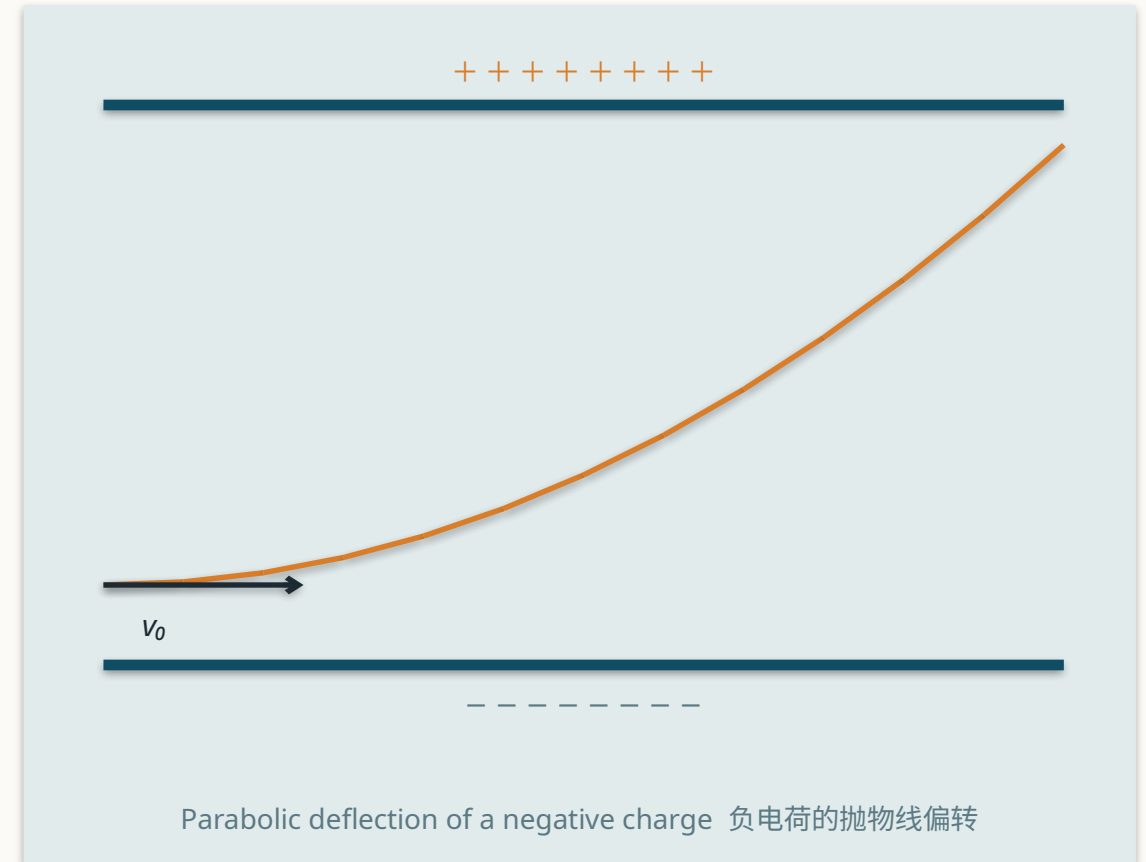
- Field lines are **parallel and evenly spaced** 平行等间距
- Derivation 推导: moving charge q across the gap, $\Rightarrow$
- E in $\text{V m}^{-1} \equiv \text{N C}^{-1}$ 两种单位等价
- Edge effects ignored — field is uniform in the central region 中央区域匀强



Charged Particles in Uniform Fields

带电粒子在匀强电场中的运动

- Force is **constant** $\Rightarrow$ constant acceleration (类重力场)
- Particle entering **parallel** to the field: speeds up / slows down in a straight line 平行入射：匀加 / 减速直线
- Particle entering **perpendicular** to the field: parabolic path 垂直入射：抛物线
- Horizontal: uniform velocity ; Vertical: uniform acceleration — just like projectile motion 类抛体运动
- Positive and negative charges deflect in **opposite** directions 正负电荷偏转方向相反



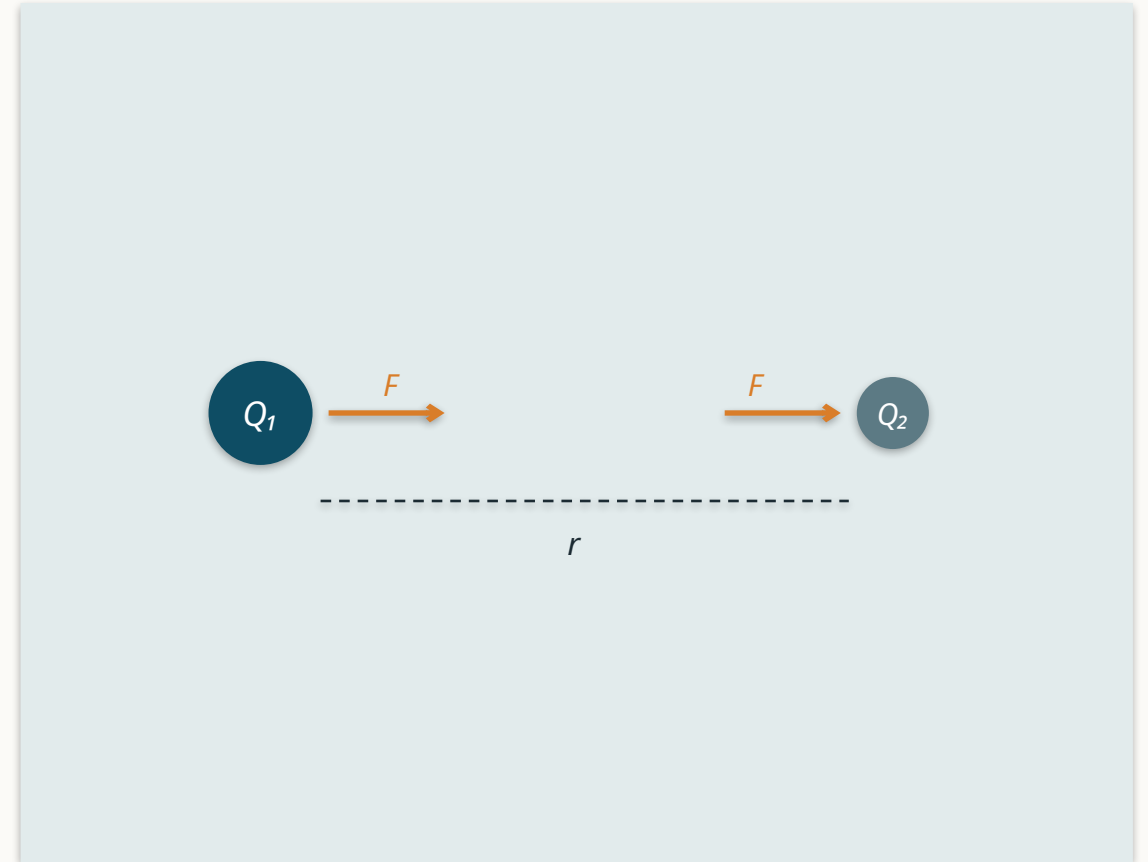
Coulomb's Law

库仑定律

$$F = \frac{Q_1 Q_2}{4 \pi \epsilon_0 r^2}$$

真空中的点电荷相互作用力

- permittivity of free space 真空介电常数
- **Like charges repel, unlike attract** 同号相斥异号相吸（与万有引力不同！）
- Inverse-square law — same form as Newton's law of gravitation 平方反比
- Charge on a spherical conductor acts as a point charge at its centre 球导体电荷等效于球心点电荷



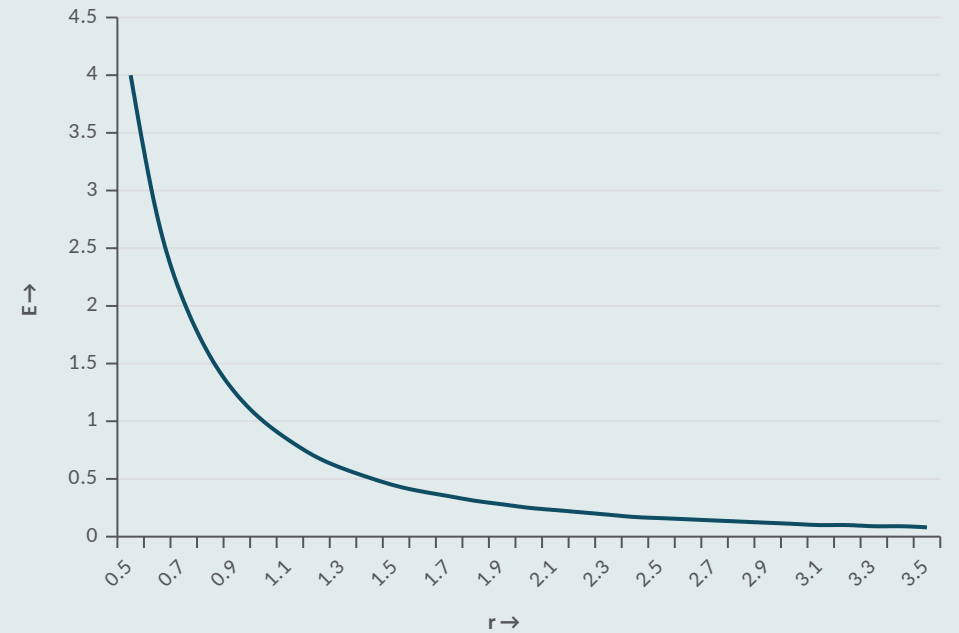
Field of a Point Charge

点电荷的电场强度

- From Coulomb's law and : 由库仑定律与场强定义

$$E = \frac{Q}{4 \pi \epsilon_0 r^2} \quad \text{radial field 径向电场}$$

- Direction: **away from +, towards -** 正电荷向外，负电荷向内
- : falls rapidly with distance 平方反比衰减
- Applies outside a charged sphere (charge acts at centre) 球外等效球心点电荷



E against r for a point charge 平方反比曲线

Electric Potential

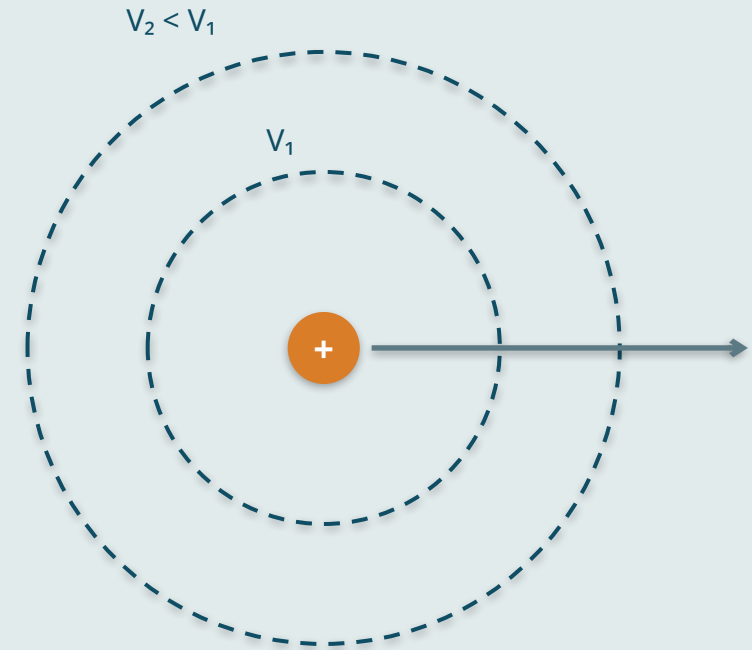
电势

Electric potential V at a point: the work done per unit positive charge in bringing a small test charge **from infinity** to that point.

把单位正电荷从无穷远移到该点所做的功。

$$V = \frac{Q}{4\pi\epsilon_0 r} \quad E_p = qV \quad \text{标量} \cdot \text{单位 volt}$$

- V is a **scalar** — add potentials algebraically 标量叠加
- Positive $Q \Rightarrow$ positive V ; negative $Q \Rightarrow$ negative V 正负由场源电荷决定
- Potential energy of charge q at that point: 电势能
- Equipotential surfaces $\perp$ field lines 等势面与电场线正交



Equipotentials around a point charge 点电荷的等势面（虚线）

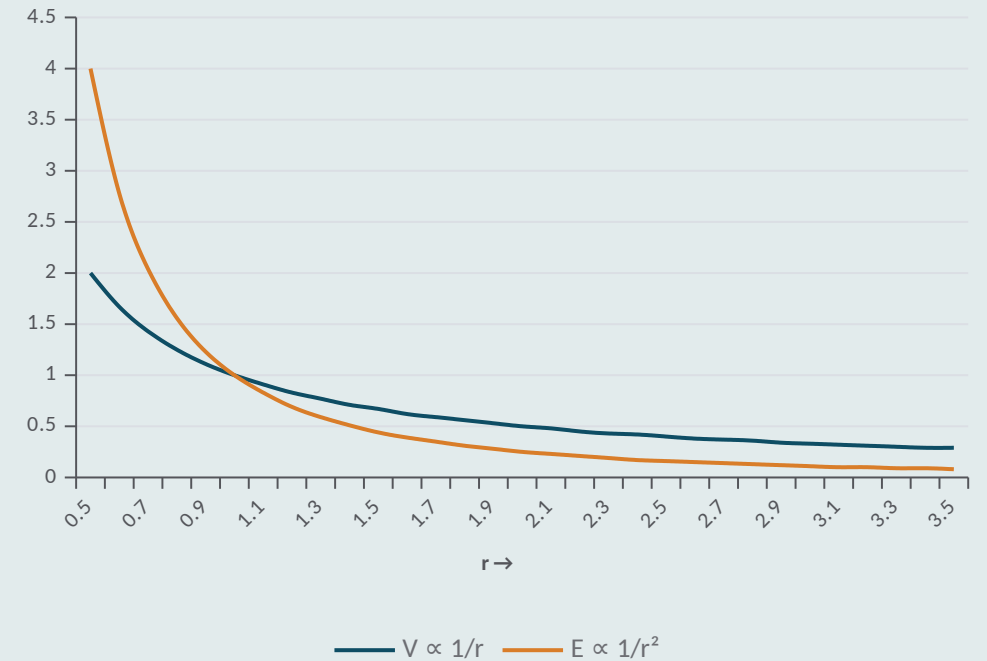
Linking Field & Potential

场强与电势的关系

- Field strength is the **negative potential gradient**: 场强 = 电势梯度的负值

$$E = - \frac{\Delta V}{\Delta r} \quad \text{uniform 特例 } E = V/d$$

- Closely spaced equipotentials $\Rightarrow$ strong field 等势面越密场越强
- Compare with gravitation: $g = -\Delta\phi/\Delta r$, $\phi = -GM/r$ 与引力场完全类比
- V is scalar, E is vector: superpose V first, then differentiate 先标量叠加再求梯度



V and E against r (normalised) 对比曲线

Worked Example 18A

例题解析

WORKED EXAMPLE 18A

Two parallel metal plates are separated by 5.0 cm. The uniform electric field between them is . Calculate the force on a charge of +5.0 nC situated mid-way between the plates.

匀强场中力与位置无关。

SOLUTION 解答

Step 1 Uniform field $\Rightarrow$ use （位置无关， mid-way 不影响结果） .

Step 2

Answer ，方向指向负极板

Worked Examples 18B & 18C

例题解析

WORKED EXAMPLE 18B

Two parallel plates are 5.0 cm apart with a p.d. of 1000 V between them. Calculate the electric field strength between the plates.

- 1.
- 2.

Answer

WORKED EXAMPLE 18C

Plates 15 mm apart, p.d. 600 V, in vacuum. An α -particle ($m = 6.7 \times 10^{-27}$ kg, $q = +3.2 \times 10^{-19}$ C) starts from rest at the + plate. Find (a) force; (b) acceleration; (c) speed at the – plate.

1. (a)
2. (b)
3. (c)

Answer (c)

Worked Examples 18D & 18E

例题解析

WORKED EXAMPLE 18D

Calculate the force between two point charges, each of $1.4 \mu\text{C}$, which are 6.0 cm apart in a vacuum. ()

- 1.
- 2.

Answer , 同号相斥 repulsive

WORKED EXAMPLE 18E

In a simplified hydrogen atom, the electron is from the proton (). Calculate the field strength of the proton at this distance.

- 1.
- 2.

Answer , 径向向外

Class Practice

随堂练习

Q1 Two plates 15 mm apart have a p.d. of 750 V. The force on a small charged sphere between them is . Calculate (a) the field strength; (b) the charge on the sphere.

Q2 Calculate the force on an α -particle (charge $+3.2 \times 10^{-19}$ C) placed from a gold nucleus (charge $+1.3 \times 10^{-17}$ C) in a vacuum. 库仑定律计算。

Q3 A Van de Graaff dome has radius 18 cm. The maximum field at its surface is . Treating the charge as concentrated at the centre, calculate the charge on the dome.

Answers Q1 (a) (b) 2.4 pC Q2 Q3 (72 nC)

Summary

本章小结

KEY FORMULAS 核心公式

$$\mathbf{E} = \frac{\mathbf{F}}{q}$$

field strength 场强

$$\mathbf{E} = \frac{\Delta V}{\Delta d}$$

uniform field 匀强场

$$F = \frac{Q_1 Q_2}{4 \pi \epsilon_0 r^2}$$

Coulomb 库仑定律

$$\mathbf{E} = \frac{Q}{4 \pi \epsilon_0 r^2}$$

point charge 点电荷

$$V = \frac{Q}{4 \pi \epsilon_0 r}$$

potential 电势

$$E_p = q V$$

potential energy 电势能

$$\mathbf{E} = - \frac{\Delta V}{\Delta r}$$

field-potential link

$$1 \text{ eV} = 1.60 \times 10^{-19} \text{ J}$$

electronvolt 电子伏特

KEY IDEAS 核心思想

- Field lines show direction and strength; uniform field between plates 场线规则与匀强场。
- Coulomb's law mirrors Newton's law of gravitation — but charges can repel 与引力定律类比。
- V is scalar, E is vector; $E = -\Delta V/\Delta r$ 标量场与矢量场的关系。

Exam Tips

备考提示

● **E is a vector, V is a scalar:** fields superpose vectorially, potentials algebraically. 叠加规则不同。

● Force direction: positive charge follows the field, negative charge opposes it. 正顺负逆。

● Unit traps: nC, μC , cm $\rightarrow$ convert before substituting. 单位换算。

● Expect comparison questions with Topic 13: v_s , ϕ vs V . 与引力场对比是高频考法。